



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2012

**HSC ASSESSMENT
TASK #2**

**Mathematics
Extension 2**

General Instructions

- Reading time – 5 minutes.
- Working time – 120 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Answers should be in simplest exact form unless specified otherwise.
- Start each **NEW** section in a separate answer booklet.
- Each section is to be returned in a separate bundle.

Total Marks - 89

- Attempt Questions 1 - 6
- All questions are NOT of equal value.

Examiner: *A. Fuller*

Section A

Question 1 (16 marks)

(a) $\int \frac{\cos x}{1+\sin x} dx$ 1

(b) $\int \frac{\cos^2 x}{1+\sin x} dx$ 2

(c) Given $a = 3 - 4i$ and $b = 1 + i$. 5

Express the following in the form $x + iy$ where x and y are real numbers:

(i) $b - a$

(ii) $\overline{ab}$

(iii) $\frac{a}{b}$

(d) $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$. 3

(i) Given that $1 + i$ is a zero of $P(x)$, explain why $1 - i$ is also a zero of $P(x)$.

(ii) Hence, find all the zeros of $P(x)$.

(e) When $(1 + ax)^5 + (1 + bx)^5$ is expanded in ascending powers of x , 5

the expansion begins $2 + 40x + 260x^2 + \dots$

(i) Show that $a + b = 8$ and $a^2 + b^2 = 26$.

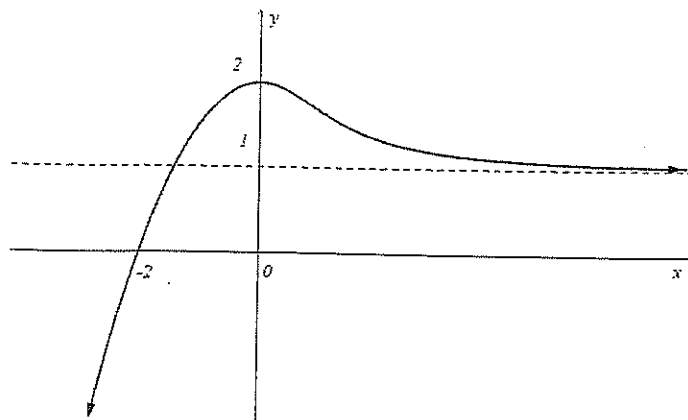
(ii) Deduce the value of ab .

(iii) Find the coefficient of x^3 .

Question 2 (16 marks)

- (a) Below is a sketch of $y = f(x)$.

6



Sketch the following on separate diagrams:

- (i) $y = f(|x|)$
 - (ii) $|y| = f(x)$
 - (iii) $y = [f(x)]^{-1}$
 - (iv) $y = 2^{f(x)}$
- (b) Plot a point A which represents the complex number z on an argand diagram 4
- given that $\operatorname{Re}(z) < 0$, $\operatorname{Im}(z) > 0$ and $|z| > 1$.
- On the same argand diagram plot the point:
- (i) B representing the complex number $\bar{z}$
 - (ii) C representing the complex number $\frac{1}{z}$
 - (iii) D representing the complex number $z(\cos \pi + i \sin \pi)$.
- (c) $\int \ln(1 + x^2) dx$. 3
- (d) Evaluate $\int_{-1}^1 (1 + x^3)^3 dx$ 3

Section B (Use a SEPARATE writing booklet)

Question 3 (14 marks)

(a) Sketch $y^2 = (x - 2)^2(x - 1)$ without using calculus. 2

(b) If $\arg(z + 1) = \frac{\pi}{6}$ and $\arg(z - 1) = \frac{2\pi}{3}$. 3

Write z in Cartesian form $(x + iy)$.

(c) The equation $x^3 - 3x^2 + ax + 8 = 0$ has roots that are in arithmetic sequence. 3

(i) Show that one of the roots is 1.

(ii) Find the value of a and solve the equation.

(d) A particle of mass 10 kg is projected vertically upwards with a velocity 6

of $u \text{ m/s}$. The resistive force is one-tenth of the square of its velocity.

Assuming that $g = 10 \text{ m/s}^2$.

(i) Show that the particle takes $\sqrt{10} \tan^{-1} \frac{u}{10\sqrt{10}}$ seconds to reach its greatest height.

(ii) Show that the greatest height is $50 \log_e \frac{1000+u^2}{1000}$ metres.

Question 4 (16 marks)

(a) (i) Show that $\int_{-a}^a f(x)dx = \int_0^a (f(x) + f(-x))dx$. 4

(ii) Hence, or otherwise, evaluate

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{1 + \sin x}$$

(b) The complex number $z = x + iy$, where x and y are real, is such that 5

$$|z - i| = \operatorname{Im}(z).$$

(i) Show that the locus of z is a parabola.

(ii) Hence, find the range of possible values for $\arg(z)$.

(c) (i) Find the values of A , B and C if 7

$$\frac{-1}{x^2(x+1)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{x+1}.$$

(ii) A particle of unit mass moves on the x -axis against a resistance numerically equal to $v^2 + v^3$, where v is its velocity. Initially the particle is travelling with velocity u , where $u > 0$.

It can be proven that when the velocity is $\frac{u}{2}$ the distance X travelled by the particle is given by $X = \ln \left(\frac{2+u}{1+u} \right)$ (Do not prove this)

(α) Prove that if T is the time taken to travel the distance X then

$$u(T + X) = 1.$$

(β) It is alleged that if the particle started at the origin then the velocity v , displacement x , and time t are related by the equation $v = \frac{u}{ux+ut+1}$.

By finding a suitable derivative, show that this is in fact correct.

Section C (Use a SEPARATE writing booklet)

Question 5 (15 marks)

(a) (i) Write $(1 + i)^n$ in modulus argument form. 6

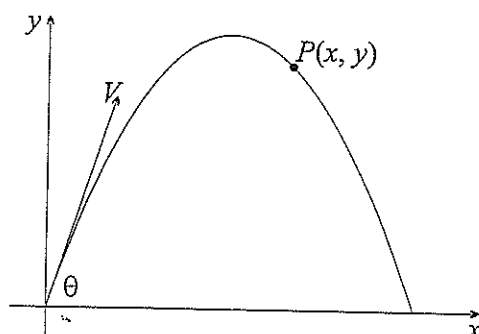
(ii) By considering the binomial expansion of $(1 + i)^n$.

Find an expression for $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots$

(iii) If n is a multiple of 8.

Show that $\binom{n}{1} + \binom{n}{5} + \binom{n}{9} + \dots = \binom{n}{3} + \binom{n}{7} + \binom{n}{11} + \dots$

(b) 9



A particle is projected at an angle of θ to the horizontal with velocity V . (Assume that there is no air resistance) and take gravity to be g . At time t , let x and y be the horizontal and vertical displacements respectively.

- (i) Derive the equations of motion in the horizontal and vertical directions in terms of t .
- (ii) It is known that at some time t during its flight, the x and y displacements of the particle are equal and the direction of motion is inclined at 45° to the downward vertical. The position of the particle at this time is marked P in the diagram above. Use this information to show that $\tan \theta = 3$.
- (iii) Hence, find the range of the particle in terms of V and g .
- (iv) If the speed of projection may be varied but the particle must not rise more than H above the ground. Find the maximum range in terms of H .

Question 6 (12 marks)

- (a) Show that the polynomial $P(x) = x^n - x^{n-1} - 1$, where $n > 1$ 3
cannot have a repeated real root.
- (b) “Words” are to be formed from the letters of the word 9

FUNDAMENTAL

- (i) Using all of the eleven letters. How many different “words”
are possible if:
- (α) there is no restriction
 - (β) it must start and end with the same letter
 - (γ) **F U N** must appear together in that order
 - (δ) the same letter must not appear next to itself?
- (ii) If I am to select five letters to form a “word”. How many different five letter
“words” are possible?

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2012 Assessment Task 2 Ex+2

Section A

Q1. (a) $\int \frac{\cos x}{1+\sin x} dx$ Let $u = \sin x$
 $du = \cos x dx$
 $= \int \frac{du}{1+u}$
 $= \ln(1+u) + C$
 $= \ln(1+\sin x) + C$ ✓ (1)

(b) $\int \frac{\cos^2 x}{1+\sin x} dx$
 $= \int \frac{1-\sin^2 x}{1+\sin x} dx = \int \frac{(1-\sin x)(1+\sin x)}{1+\sin x} dx$ ✓
 $= \int (1-\sin x) dx$ (2)
 $= x + \cos x + C$ ✓

(c) $a = 3-4i$, $b = 1+i$.

(i) $b-a = -2+5i$ ✓

(ii) $ab = 7-i$ ✓ $\therefore \bar{ab} = \underline{7+i}$ ✓ (5)

(iii) $\frac{a}{b} = \frac{3-4i}{1+i}$

$= \frac{3-4i}{1+i} \times \frac{1-i}{1-i}$ ✓

$= \frac{-1-7i}{1+1} = \frac{-1-7i}{2} = \underline{-\frac{1}{2} - \frac{7}{2}i}$ ✓

1.(d) $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$

(i) By the conjugate root theorem, if $P(x)$ has real coefficients and $(a+bi)$ is a root, then so is $(a-bi)$.
 $\therefore$ If $(1+i)$ is a zero so is $(1-i)$ since $P(x)$ has real coefficients. ✓

(ii) $(x-1-i)(x-1+i)$ is a factor of $P(x)$

$\Rightarrow (x^2-2x+2)$ is a factor ✓

By division $\begin{array}{r} x^2+x-2 \\ x^4-x^3-2x^2+6x-4 \\ \underline{x^4-2x^3+2x^2} \\ x^3-4x^2+6x \\ \underline{x^3-2x^2+2x} \\ -2x^2+4x-4 \\ \underline{-2x^2+4x-4} \\ 0 \end{array}$ (3)

for $q(x) = x^2+x-2 = (x+2)(x-1)$
 $x = \frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2}$
 $x = -2 \text{ or } 1$ ✓

$\therefore$ zeros of $P(x)$ are $x = \underline{-2, 1, (1+i), (1-i)}$

$$1(e) (1+ax)^5 + (1+bx)^5 = 2 + 40x + 260x^2 + \dots$$

$$\text{For } (1+ax)^5, T_{r+1} = {}^5C_r (ax)^r$$

$$\text{For } (1+bx)^5, T_{r+1} = {}^5C_r (bx)^r$$

$$\therefore T_{r+1}(\text{of sum}) = {}^5C_r (a^r + b^r) x^r$$

$$\text{Coefficient of term in } x^1 = {}^5C_1 (a+b) = 40$$

$$\therefore a+b = 8 \quad (1) \checkmark$$

$$\text{Coefficient of term in } x^2 = {}^5C_2 (a^2 + b^2) = 260$$

$$10(a^2 + b^2) = 260$$

$$a^2 + b^2 = 26 \quad (2) \checkmark$$

$$(ii) \text{ Now } (a+b)^2 = (a^2 + b^2) + 2ab = 8^2 \quad \checkmark$$

$$26 + 2ab = 64$$

from (2) and (1)

$$2ab = 38$$

$$ab = 19 \quad \checkmark$$

$$(iii) \text{ Coefficient of } x^3 = {}^5C_3 (a^3 + b^3)$$

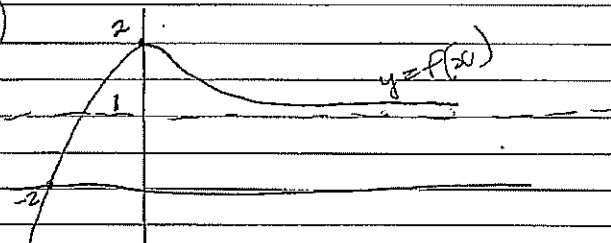
$$= 10(a+b)(a^2 - ab + b^2)$$

$$= 10 \times 8 \times (26 - 19)$$

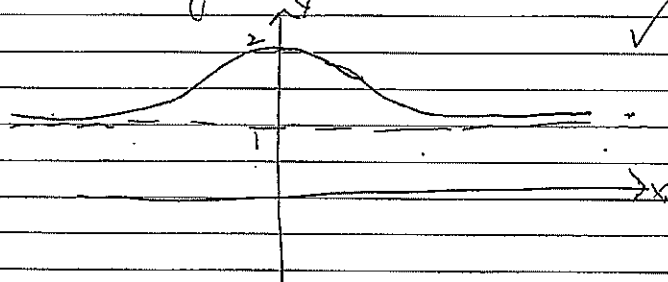
$$= 560 \quad \checkmark$$

(5)

Q2. (a)

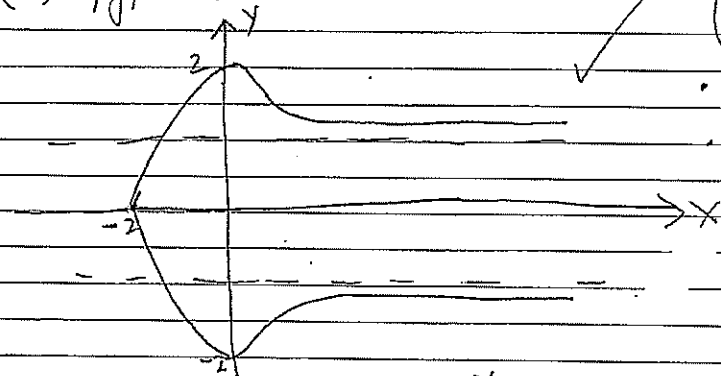


$$(i) y = f(|x|)$$



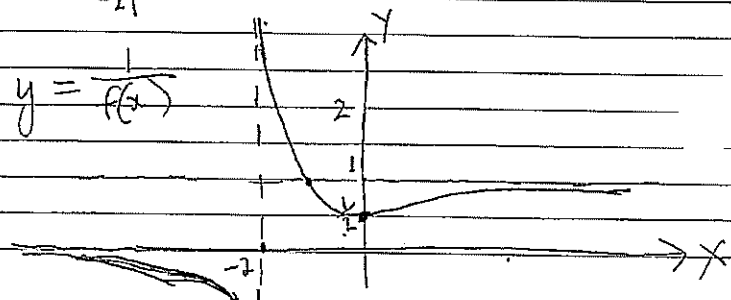
Sketch $y = f(x)$ for $x \geq 0$ only.
Then reflect in y.

$$(ii) |y| = f(x)$$



Discard section of $f(x)$ below x-axis.
Reflect leftover section in x=0.

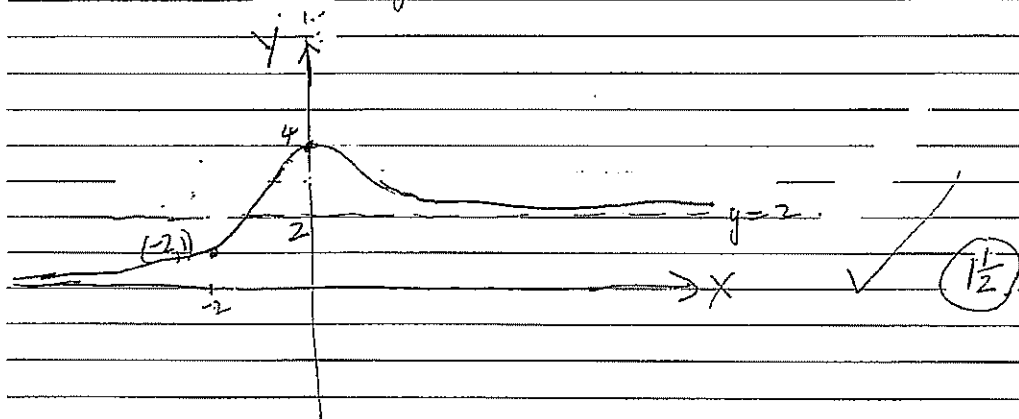
$$(iii) y = \frac{1}{f(x)}$$



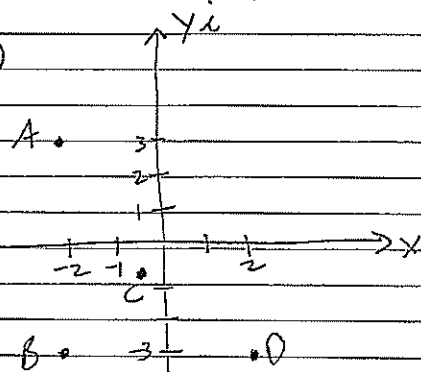
(1/2)

2(a)
(iv) $y = 2^{f(x)}$

x	-2	0	∞	$-\infty$
$f(x)$	0	2	1^+	$-\infty$
$y = 2^{f(x)}$	1	4	2^+	0^+



(b)



eg $A \rightarrow -2 + 3i = z$
 $|z| = \sqrt{4+9} = \sqrt{13}$

$B \rightarrow \bar{z} = -2 - 3i$

$C = \frac{1}{z} = \frac{-2-3i}{\sqrt{13}}$
 $= \frac{-2}{\sqrt{13}} - \frac{3}{\sqrt{13}}i$
 $(\approx -0.55 - 0.83i)$

In general

$A = -a + bi$

$B = -a - bi$

$C = \frac{-a-bi}{\sqrt{a^2+b^2}}$

$D = -z = a - bi$

$D = z \cdot \text{cis } \pi$
 $= z \cdot (-1 + 0i)$
 $= -z$
 $= 2 - 3i$

Q2(c) $\int \ln(1+x^2) dx$
 $= \int 1 \cdot \ln(1+x^2) dx$

parts $\int u dv = uv - \int v du$ where $u = \ln(1+x^2)$ $dv = 1$
 $du = \frac{2x}{1+x^2}$ $v = x$

$I = x \ln(1+x^2) - \int x \cdot \frac{2x}{1+x^2} dx$ ✓

$= x \ln(1+x^2) - 2 \int \frac{x^2}{1+x^2} dx$

$= x \ln(1+x^2) - 2 I_2$

Now $I_2 = \int \frac{x^2+1-1}{x^2+1} dx$ ✓

$= \int 1 dx - \int \frac{1}{x^2+1} dx$ (3)
 $= x - \tan^{-1}x$

$\therefore I = x \ln(1+x^2) - 2(x - \tan^{-1}x) + C$

$I = x \ln(1+x^2) - 2x + 2 \tan^{-1}x + C$ ✓

(d) $\int_{-1}^1 (1+x^3)^3 dx$

$= \int_{-1}^1 (1 + 3x^3 + 3x^6 + x^9) dx$ ✓

$= \left[x + \frac{3x^4}{4} + \frac{3x^7}{7} + \frac{x^{10}}{10} \right]_{-1}^1$ ✓ (3)

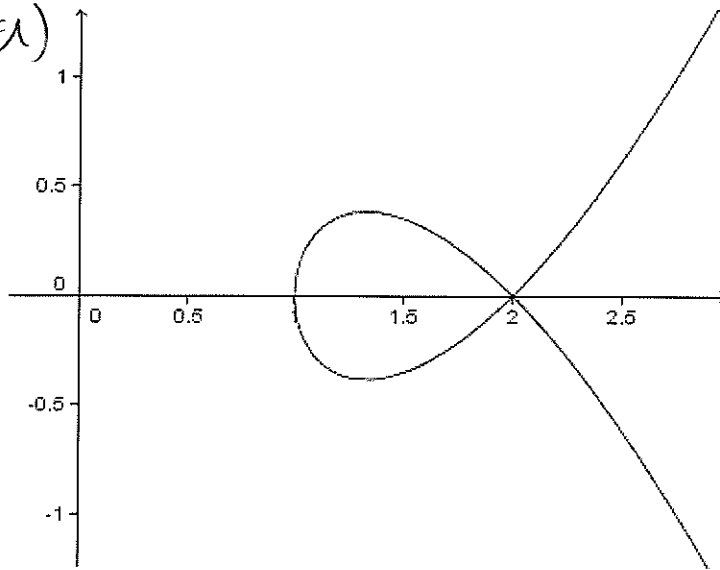
$= \left(1 + \frac{3}{4} + \frac{3}{7} + \frac{1}{10} \right) - \left(-1 + \frac{3}{4} - \frac{3}{7} + \frac{1}{10} \right)$

$= \underline{\underline{\frac{20}{7}}} = \underline{\underline{\frac{20}{7}}}$ ✓

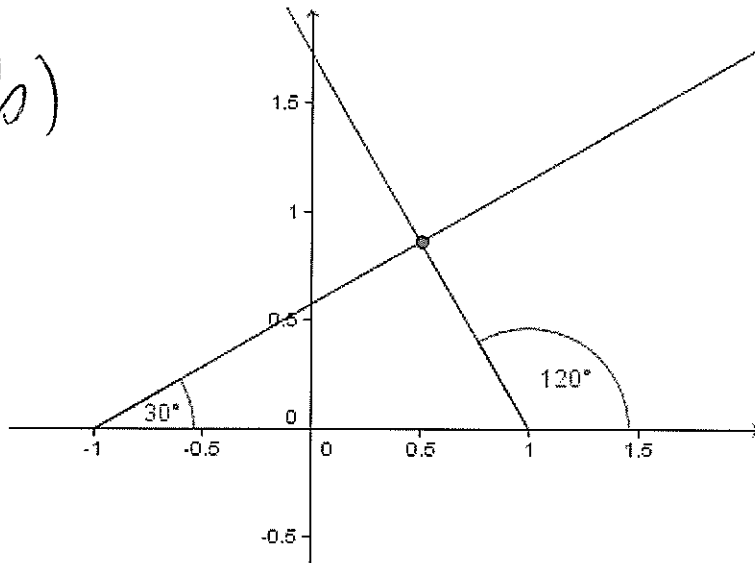
SECTION B

Q4

(a)



(b)



$$m_1 = \tan \frac{2\pi}{3}$$
$$= -\sqrt{3}$$

point (1,0)

$$y = -\sqrt{3}(x-1)$$

$$m_2 = \tan \frac{\pi}{6}$$
$$= \frac{1}{\sqrt{3}}$$

point (-1,0)

$$y = \frac{1}{\sqrt{3}}(x+1)$$

intesection

$$-\sqrt{3}(x-1) = \frac{1}{\sqrt{3}}(x+1)$$

$$-3x+3 = x+1$$

$$-4x = -2$$

$$x = \frac{1}{2}$$

$$y = -\sqrt{3}\left(\frac{1}{2}-1\right)$$

$$y = \frac{\sqrt{3}}{2}$$

$$z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(c) (i) $\alpha + \beta, \alpha, \alpha - \beta.$

$$3\alpha = -\frac{b}{a}$$

$$3\alpha = -3$$

$$\alpha = -1$$

(ii) $1 + \beta, 1, 1 - \beta.$

$$p(1) = 1^3 - 3 + a + 8 = 0$$

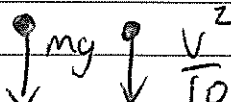
$$a = -6$$

$$\sum \alpha_i \alpha_j = \frac{c}{a}$$

$$1 + \beta + 1 - \beta + 1 - \beta^2 = -6$$

$$\beta = \pm 3$$

$$x = 4, 1, -2.$$

(d) 

$$10\ddot{x} = -100 - \frac{v^2}{10}$$

$$100 \frac{dv}{dt} = -1000 - v^2$$

$$\frac{100}{1000 + v^2} \frac{dv}{dt} = -1$$

$$100 \int_u^0 \frac{dv}{(10\sqrt{10})^2 + v^2} = - \int_0^t dt$$

$$100 \left[\frac{1}{10\sqrt{10}} \tan^{-1} \frac{v}{10\sqrt{10}} \right]_u^0 = -t$$

$$t = \sqrt{10} \tan^{-1} \left(\frac{u}{10\sqrt{10}} \right).$$

$$(ii) \quad 10 \ddot{x} = -100 - \frac{v^2}{10}$$

$$100 \ddot{x} = -1000 - v^2$$

$$100 \frac{dv}{dx} \frac{dx}{dt} = -1000 - v^2$$

$$\frac{100v}{1000 + v^2} \frac{dv}{dx} = -1$$

$$50 \int_u^0 \frac{2v}{1000 + v^2} dv = - \int_0^x dx$$

$$50 \left[\ln(1000 + v^2) \right]_u^0 = -x$$

$$x = 50 \ln \left(\frac{1000 + u^2}{1000} \right)$$

Q4
(a) (i) $LHS = F(a) - F(-a)$

where $F' = f$

$$RHS = \int_0^a f(x) dx + \int_0^a f(-x) dx.$$

$$= F(a) - F(0) - F(-a) + F(0)$$

$$= F(a) - F(-a)$$

$$= LHS.$$

(ii) $\int_{-\pi/4}^{\pi/4} \frac{dx}{1+\sin x} = \int_0^{\pi/4} \frac{1}{1+\sin x} dx + \int_0^{\pi/4} \frac{1}{1-\sin x} dx$

Since
 $\sin(-x) = -\sin x$

$$= \int_0^{\pi/4} \frac{2}{1-\sin^2 x} dx$$

$$= \int_0^{\pi/4} \frac{1}{\cos^2 x} dx.$$

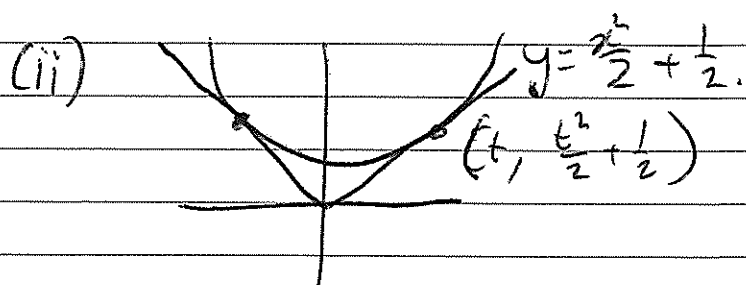
$$= \int_0^{\pi/4} \sec^2 x dx$$

$$= 2.$$

$$(b)(i) |x + i(y-1)| = y$$

$$x^2 + (y-1)^2 = y^2$$

$$y = \frac{x^2}{2} + \frac{1}{2}$$



$$\frac{dy}{dx} = x.$$

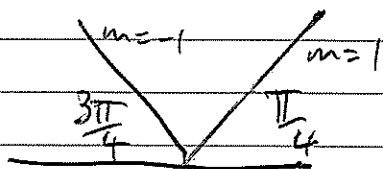
let $m = t$. point $(0, 0)$.

$$y - (\frac{t^2}{2} + \frac{1}{2}) = t(x - t)$$

$$-\frac{t^2}{2} - \frac{1}{2} = -t^2$$

$$t^2 = 1$$

$$t = \pm 1$$



$$\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}.$$

(c) (i)

$$-1 \equiv A(x+1) + Bx(x+1) + Cx^2$$

When $x = -1$

$$-1 = C$$

x^2

$$Bx^2 + Cx^2 = 0x^2$$

When $x = 0$

$$A = -1$$

$$B + C = 0$$

$$B - 1 = 0$$

$$B = 1$$

$$\frac{-1}{x^2(x+1)} = \frac{-1}{x^2} + \frac{1}{x} + \frac{1}{x+1}$$

(ii) $\ddot{x} = -v^2 - v^3$

$$\frac{dv}{dt} = -(v^2 + v^3)$$

$$\frac{dt}{dv} = -\frac{1}{v^2(v+1)}$$

$$= -\frac{1}{v^2} + \frac{1}{v} + \frac{-1}{v+1}$$

$$t = \frac{1}{v} + \ln v - \ln(v+1) + C$$

when $t = 0$ $v = u$

$$C = -\frac{1}{u} + \ln\left(\frac{u+1}{u}\right)$$

$$t = \frac{1}{v} + \ln v - \ln(v+1) - \frac{1}{u} + \ln\left(\frac{u+1}{u}\right)$$

$$\text{when } t = T \quad v = \frac{u}{2}$$

$$T = \frac{2}{u} + \ln\left(\frac{\frac{u}{2}}{\frac{u}{2}+1}\right) - \frac{1}{u} + \ln\left(\frac{u+1}{u}\right)$$

$$T = \frac{1}{u} + \ln\left(\frac{u+1}{u+2}\right) \quad \text{But } X = \ln\left(\frac{2+u}{1+u}\right)$$

$$T = \frac{1}{u} - \ln\left(\frac{2+u}{1+u}\right)$$

$$T = \frac{1}{u} - X$$

$$u(T+X) = 1$$

$$(ii) \quad v = u(ux+ut+1)^{-1}$$

$$\frac{dv}{dt} = -u(ux+ut+1)^{-2} \left(u \frac{dx}{dt} + u\right)$$

$$\ddot{x} = \frac{-u}{(ux+ut+1)^2} (uv+u)$$

$$= \frac{-u^2(v+1)}{\left(\frac{u}{v}\right)^2}$$

$$= -v^2(v+1)$$

$$= -v^3 - v^2$$

SECTION C Q5

$$(a) \quad (i) \quad (1+i)^n = (\sqrt{2} \cos \frac{\pi}{4})^n \\ = (\sqrt{2})^n \cos \frac{n\pi}{4} \quad (A)$$

$$(ii) \quad (1+i)^n = \binom{n}{0} + \binom{n}{1}i + \binom{n}{2}i^2 + \binom{n}{3}i^3 + \binom{n}{4}i^4 + \dots \\ = \binom{n}{0} + \binom{n}{1}i - \binom{n}{2} + \binom{n}{3}i + \binom{n}{4} - \dots \\ = \left(\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots \right) + i \left(\binom{n}{1} - \binom{n}{3} + \dots \right)$$

now real part is $\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots$
which is $\left| (\sqrt{2})^n \cos \frac{n\pi}{4} \right|$ from (A)

(iii) If n is a multiple of 8. say $8m$

$$\text{In (A)} \quad (\sqrt{2})^n \cos \frac{8m\pi}{4}$$

$$= (\sqrt{2})^n \left[\cos 2m\pi + i \sin 2m\pi \right]$$

$$= (\sqrt{2})^n [1 + 0]$$

$\therefore$ imaginary part is zero.

$$\text{i.e.} \quad \binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \binom{n}{7} + \binom{n}{9} - \binom{n}{11} + \dots = 0$$

$$\text{i.e.} \quad \left| \binom{n}{1} + \binom{n}{5} + \binom{n}{9} + \dots = \binom{n}{3} + \binom{n}{7} + \binom{n}{11} + \dots \right|$$

$$(b)(i) \quad \boxed{\ddot{x} = 0}$$

$$\dot{x} = c_1$$

$$\therefore \boxed{\dot{x} = v \cos \theta}$$

$$x = vt \cos \theta + c_3$$

$$\text{clearly } c_3 = 0$$

$$\therefore \boxed{x = vt \cos \theta}$$

$$\boxed{\ddot{y} = -g}$$

$$\dot{y} = -gt + c_4$$

$$\therefore \boxed{\dot{y} = -gt + v \sin \theta}$$

$$y = -\frac{1}{2}gt^2 + vt \sin \theta + c_4$$

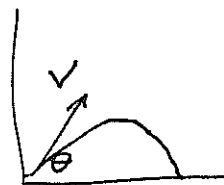
$$c_4 = 0$$

$$\therefore \boxed{y = -\frac{1}{2}gt^2 + vt \sin \theta}$$

$$t=0$$

$$x=0$$

$$y=0$$



$$\dot{y} = v \sin \theta$$

$$\dot{x} = v \cos \theta$$

$$(ii). \text{ now } x = y$$

$$\therefore vt \cos \theta = -\frac{1}{2}gt^2 + vt \sin \theta$$

$$\text{OR } \boxed{v \cos \theta = -\frac{1}{2}gt + v \sin \theta} \quad (A) \quad (\text{NB } t \neq 0)$$

$$\& \quad \dot{x} = -\dot{y}$$

$$\therefore \boxed{v \cos \theta = gt - v \sin \theta} \quad (B)$$

$$(A) - (B)$$

$$\boxed{0 = -\frac{3}{2}gt + 2v \sin \theta} \quad (C) \Rightarrow \boxed{v \sin \theta = \frac{3}{4}gt}$$

$$(A) + (B)$$

$$2v \cos \theta = \frac{1}{2}gt$$

$$\text{ie } \boxed{v \cos \theta = \frac{1}{4}gt} \quad (D)$$

$$\text{From (C) \& (D)}$$

$$\frac{v \sin \theta}{v \cos \theta} = \frac{\frac{3}{4}gt}{\frac{1}{4}gt}$$

$$\therefore \boxed{\tan \theta = 3}$$

Q6. (a) $P(x) = x^n - x^{n-1} - 1 \quad ; \quad n > 1.$

$$\& P'(x) = nx^{n-1} - (n-1)x^{n-2}$$

For a repeated root $\alpha \quad P(\alpha) = P'(\alpha) = 0$

Consider $P'(\alpha) = n\alpha^{n-1} - (n-1)\alpha^{n-2} = 0.$

$$\text{ie } \alpha^{n-2} [n\alpha - (n-1)] = 0$$

$$\alpha \neq 0. \text{ as } P(\alpha) = -1$$

$$\therefore \alpha = \frac{n-1}{n}.$$

Assume $P(\alpha) = 0$

$$\text{ie. } \left(\frac{n-1}{n}\right)^n - \left(\frac{n-1}{n}\right)^{n-1} - 1 = 0$$

$$\text{ie. } \left(\frac{n-1}{n}\right)^{n-1} \left[\frac{n-1}{n} - 1 \right] = 1$$

$$\left(\frac{n-1}{n}\right)^{n-1} \left(-\frac{1}{n}\right) = 1$$

This is impossible

$$\text{since } \left(\frac{n-1}{n}\right)^{n-1} > 0 \quad [n > 1]$$

$$\& -\frac{1}{n} < 0$$

$\therefore$ LHS is negative

$\therefore$ Contradiction

$$P(\alpha) \neq 0$$

$\therefore$ no repeated roots.

Q6 (b) (1) (a) $\frac{11!}{2! \times 2!} = \underline{9,979,200}$

(b) $\frac{9!}{2} \times 2 = \underline{362,880}$

(c) $\frac{9!}{2} = \underline{181,440}$

(d) $\frac{11!}{2! \times 2!} - \frac{10!}{2} - \frac{10!}{2} + 9! = \underline{6,713,280}$

(11) Look at cases.

$$7 \times \frac{5!}{2! \times 2!} + \binom{7}{2} \times \frac{5!}{2!} \times 2 + \binom{7}{3} \times \frac{5!}{2!} \times 2$$

$$+ \binom{7}{3} \times 5! + \binom{7}{4} \times 5! \times 2 + 7P_5$$

$$= 210 + 2520 + 420 + 420 + 840 + 2520$$

$$= \underline{22,050}$$